

OJAS PATIL

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SUMMARY

Results-driven **Computer Science student at Arizona State University** passionate about **building scalable, production-ready systems**. Experienced in **Python, Java, and C++** with a strong foundation in **data structures, algorithms, and object-oriented design**. Proven ability to solve complex problems, work through ambiguity, and take ownership of impactful technical deliverables end-to-end.

EDUCATION

BS in Computer Science, Minor in Business Graduating May 2028
3.88
Arizona State University, Tempe, AZ
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Calculus

SKILLS

Programming Languages: Java, Python, C, C++, HTML, CSS, JavaScript, MySQL
Technologies: Flask, TensorFlow, Pandas, NumPy, Matplotlib, Scikit-learn, Pyttsx3, REST APIs
Additional Skills: Git, Github, Linux, TinkerCAD, 3D Printing Tools

PROJECTS

- URLS | Python, Flask, SQLite, djb2 Hashing

March 2026

- Engineered a **collision-resistant URL shortener** using **djb2 hashing with base62 encoding**, supporting 3.5 trillion unique short codes via a RESTful API. [Repo]
 - Designed a **scalable object-oriented backend** with B-tree indexed SQLite schema and click analytics tracking timestamps, IP addresses, and user agents.
 - Documented a **Redis caching + Nginx load balancing** scaling roadmap to support 1M+ URLs under production traffic.
- VeriQ | Google ADK, Gemini, Python

November 2025

- Built a **5-agent LLM-powered** data quality pipeline for KPI monitoring and incident triage. [Repo]
 - Detected anomalies via **14-day rolling z-score** $z \geq 2.0$ with **schema + changelog** root-cause context.
 - Generated a structured **PDF data quality report** per run, summarizing outliers, suspected causes, and next-step actions for faster debugging.
- Portfolio Checker Website | Python, Flask, HTML/CSS, JavaScript, YahooFinance API

October 2024 – June 2025

- Built a **real-time portfolio tracking web app** to visualize stock performance using Flask and YahooFinance API. [Repo]
 - Designed a **responsive dashboard** with dynamic charts, improving data interpretation efficiency by **40%**.
 - Automated daily valuation updates and analytics using Pandas and RESTful endpoints.
- GCSP | MicroGuardian – Precision Fire-Fighting Drone Swarm

January 2025 – May 2025

- Designed and simulated a **VTOL drone swarm** that reduced wildfire spread by **30%** and response time by **70%**. [Repo]
 - Integrated **graphene super-capacitor research** to improve energy efficiency and flight duration.
 - Presented the project as part of the **Grand Challenges Scholars Program (GCSP)** on security.
- EPICS | 3D Printed Dental Devices | 3D Printing, Material Science, TinkerCAD

August 2024 – December 2024

- Led the **3D design and prototyping** of dental crowns in TinkerCAD for community-based healthcare innovation.
 - Conducted **5+ lab durability tests**, increasing tensile strength of models by **20%** using advanced materials.
- Voice Assistant Chatbot | Python, Pyttsx3, Spotify API, Weather API

March 2024 – June 2024

- Engineered a **voice-controlled assistant** integrating real-time Spotify and weather data.
 - Optimized NLP pipeline to achieve **90%+ command recognition accuracy**.

EXPERIENCE

Camp Counselor (C2), E2 Camp - ASU, July 2025 - August 2025

- Mentored 50+ incoming freshmen, improving their adjustment to college through leadership activities.
- Led the **Maroon Group** to a **Gold Medal** in team competitions by fostering collaboration and engagement.

CERTIFICATIONS & OTHER EXPERIENCES

Prompt Engineering & Programming with OpenAI – Columbia+ July 2025

- Built **AI-driven apps** using **OpenAI API, LangChain, and LlamaIndex** for automation and intelligent querying. [Credentials]

Google AI Essentials V1 – Coursera July 2025

- Gained proficiency in **responsible AI**, generative tools, and data-driven decision making. [Credentials]